

MIPI CSI-2 IP User Guide

Updated for Quartus® Prime Design Suite: **26.1.1**

IP Version: **4.0.0**



Online Version



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1. About the MIPI CSI-2 IP

The MIPI Camera Serial Interface 2 (MIPI CSI-2) IP is a high-speed protocol IP that receives or transmits MIPI CSI-2 video between image sensors, processors, and bridging applications (RX, TX, and passthrough modes). This IP is compatible with the MIPI D-PHY IP.

1.1. MIPI CSI-2 IP Features

- MIPI CSI-2 Protocol Layer (transmitter and receiver)
- 1, 2, 4, and 8 D-PHY lanes
- 1, 2, and 4 pixels in parallel
- Support for data formats RAW6, RAW7, RAW8, RAW10, RAW12, RAW14, RAW16, RAW20, RAW24, RGB444, RGB555, RGB565, RGB666, RGB888, YUV420 8-bit, 10-bit, and 8-bit legacy modes, and YUV422 8-bit and 10-bit
- Avalon® memory-mapped interface for memory access
- AMBA AXI4-Stream interface for video data streaming
- MIPI PHY-Protocol Interface (PPI) compatible with MIPI D-PHY IP
- Passthrough mode for receiver-to-transmitter bridging applications, bypassing pixel decode and encode

1.2. Device Family Support for MIPI CSI-2 IP

The following terms define IP support levels for IP device families and state the level of support available in the current release:

- **Advance support**—the IP is available for simulation and compilation for this device family. Timing models include initial engineering estimates of delays based on early post-layout information. The timing models are subject to change as silicon testing improves the correlation between the actual silicon and the timing models. You can use this IP for system architecture and resource utilization studies, simulation, pinout, system latency assessments, basic timing assessments (pipeline budgeting), and I/O transfer strategy (datapath width, burst depth, I/O standards tradeoffs).
- **Preliminary support**—the IP is verified with preliminary timing models for this device family. The IP meets all functional requirements, but might still be undergoing timing analysis for the device family. It can be used in production designs with caution.
- **Final support**—the IP is verified with final timing models for this device family. The IP meets all functional and timing requirements for the device family and can be used in production designs.

Table 1. MIPI CSI-2 IP Device Family Support

Device Family	Support Level
Agilex® 3	Preliminary
Agilex 5 D-series	Advance
Agilex 5 E-series	Preliminary
Other device families	No support

1.3. Recommended Speed Grades for MIPI CSI-2 IP

The following table shows the recommended speed grade when axi4s_clk runs at certain clock frequencies. To calculate the required clock frequency, refer to *Transmitter Clock Requirements*.

Table 2. Recommended Speed Grades

Device	axi4s_clk Frequency (MHz)	Speed Grades
Agilex 5 D-Series	<450	-3, -2, -1
	450 to 500	-2, -1
Agilex 5 E-Series (Group A)	<400	-3, -2, -1
	400 to 500	-2, -1
Agilex 5 E-Series (Group B)	<250	-6, -5, -4
	250 to 300	-5, -4
	300 to 400	-4
Agilex 3	<250	-7, -6
	250 to 345	-6

MIPI data rates are supported up to the maximum rate of the FPGA on all speed grades:

- 2500Mbps per lane on Agilex 3 and Agilex 5 E-Series Group B FPGAs.
- 3500Mbps per lane on Agilex 5 D-Series and Agilex 5 E-Series Group A FPGAs.

Related Information

- [MIPI CSI-2 Transmitter Clock Requirements](#) on page 26
- [D-PHY PPI Clock Frequencies](#) on page 26

1.4. MIPI CSI-2 IP Performance and Resources

Shows the typical device resources for selected configurations using the Quartus® Prime Pro Edition software. The number of ALMs and logic registers are rounded up to the nearest 50. The number stated for M20K memory blocks includes no rounding.

Changing pixels-in-parallel or data type has little impact on resources, though selecting multiple data types does increase logic.

The device type (Agilex 3, Agilex 5 E-Series, or Agilex 5 D-Series) makes no difference.

Table 3. CSI-2 Receiver Performance and Resources

Turning off ECC and CRC checking, and removing the **Control and status registers** reduces logic to around 35% of these values.

Passthrough reduces logic to around 75%.

Configuration				ALMs	Logic Registers	M20K Memory Blocks
Data Lanes	Pixel-in-Parallel	Number of Interfaces	Data Type			
1	2	1	RAW8	2,550	4,900	6
1	2	4	RAW8	3,800	7,050	18
2	2	1	RAW8	2,650	5,150	8
2	2	4	RAW8	3,950	7,400	20
4	2	1	RAW8	3,450	6,350	12
4	2	4	RAW8	4,650	8,650	24
8	2	1	RAW8	5,200	8,950	20
8	2	4	RAW8	6,450	11,300	32

Table 4. CSI-2 Transmitter Performance and Resources

Turning off ECC and CRC insertion (which is against the specification and hence not recommended), and turning off the **Control and status registers** reduces logic to around 70% of these values.

Passthrough reduces logic to around 25%.

Configuration				ALMs	Logic Registers	M20K Memory Blocks
Data Lanes	Pixel-in-Parallel	Number of Interfaces	Data Type			
1	2	1	RAW8	2,050	3,750	8
1	2	4	RAW8	3,800	6,450	11
2	2	1	RAW8	2,050	4,150	8
2	2	4	RAW8	3,750	7,250	11
4	2	1	RAW8	2,550	4,650	8
4	2	4	RAW8	4,350	7,250	11
8	2	1	RAW8	4,200	6,900	10
8	2	4	RAW8	6,000	9,300	13

1.5. Release Information for MIPI CSI-2 IP

Table 5. Release information for the MIPI CSI-2 IP

Item	Description
IP Version	4.0.0
Quartus Prime Pro Edition Version	26.1.1
Release Date	2026.08.10
Ordering Code	IP-MIPI-DSI-CSI

1.6. Licensing the IP

The IP is included with the Quartus Prime Pro Edition software and is available in the IP Catalog and Platform Designer. A specific IP license for the IP is required, which is available from the Altera Self-Service Licensing Center.

After logging into the Licensing Center:

- Select **Sign up for Evaluation or No-Cost Licenses**.
- Add the MIPI DSI-2/MIPI CSI-2 IP license (IP-MIPI-DSI-CSI).
- Follow the prompts to generate a license for your system.

Note that the IP uses some components from the Video and Vision Processing (VVP) Suite internally. As a result, any design that uses the MIPI CSI-2 shows the VVP license in the list of licensed IP in the system. However, a VVP license is not required to build the MIPI CSI-2 IP.

2. MIPI CSI-2 IP Interfaces

2.1. MIPI CSI-2 IP Clocks

Table 6. MIPI CSI-2 IP Clocks

Clock	Description
axi4s_clk	AXI4-Stream video clock. The IP uses this clock when the independent control clock parameter is disabled. It is used for AXI4-Stream video interfacing and processing, and control and status registers. Your system should generate this clock.
slave_clock	Clock for Avalon control and status register interface when the independent control clock parameter is enabled. When this option is disabled, the port is not present.
ls_clk	This clock is for RX only. When MIPI continuous clock is disabled, this port is present to ensure packet processing can be completed. This must be the same frequency as the MIPI receive word clock: rx_word_clk_hs_ck.
rx_word_clk_hs_ck	High-speed receive word clock – clock lane. The IP uses this clock to synchronize signals in the high-speed receive clock domain. ⁽¹⁾
rx_word_clk_hs_d<lane>	High-speed receive word clock – data lanes. The IP uses this clock to synchronize signals in the high-speed receive clock domain. ⁽¹⁾
tx_word_clk_hs_ck	High-speed transmit word clock – clock lane. The IP uses this clock to synchronize signals in the high-speed transmit clock domain. ⁽¹⁾
tx_word_clk_hs_d<lane>	High-speed transmit word clock – data lanes. The IP uses this clock to synchronize signals in the high-speed transmit clock domain. ⁽¹⁾

2.2. MIPI CSI-2 IP Resets

Table 7. MIPI CSI-2 IP Resets

Reset	Associated Clock Domain	Description
axi4s_rst	axi4s_clk	Asserting this reset triggers a reset to all the blocks in axi4s_clk clock domain.
slave_reset	slave_clock	Reset for the Avalon control and status register interface when the independent control clock parameter is enabled. When this option is disabled, the port is not present.
ls_rst	ls_clk	Reset for the packet receive logic when the MIPI continuous clock option is disabled (RX only).
rx_srst_n_ck	rx_word_clk_hs_ck	This signal is part of receiver PPI output signals, which should be connected to MIPI D-PHY IP.
continued...		

⁽¹⁾ The D-PHY receiver IP generates this clock, which connects to the MIPI CSI-2 IP via the PPI.

Reset	Associated Clock Domain	Description
rx_srst_n_d<lane>	rx_word_clk_hs_d<lane>	This signal is part of receiver PPI output signals, which should be connected to MIPI D-PHY IP.
tx_srst_n_ck	tx_word_clk_hs_ck	This signal is part of transmitter PPI input signals, which should be connected to MIPI D-PHY IP.
tx_srst_n_d<lane>	tx_word_clk_hs_d<lane>	This signal is part of transmitter PPI input signals, which should be connected to MIPI D-PHY IP.

2.3. MIPI CSI-2 IP User Interfaces

2.3.1. Receiver PPI Signals

The interface between the MIPI CSI-2 IP and the Altera MIPI D-PHY IP uses the PPI defined in the MIPI D-PHY and CSI-2 standards. This interface carries the received MIPI data stream to be decoded by the CSI-2 Receiver.

Table 8. Receiver PPI Signals

Signal	Width	Direction	Description
Clock Lane High Speed Receive <ppi_width>=16			
rx_data_width_hs_ck	2	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 95 and 97, 101 High Speed Receive on Slave Side, 8/16-bit Bus, LP Mode.</i>
rx_data_hs_ck	ppi_width	Input	
rx_valid_hs_ck	ppi_width/8	Input	
rx_active_hs_ck	1	Input	
rx_sync_hs_ck	1	Input	
rx_clk_active_hs_ck	1	Input	Refer to <i>MIPI D-PHY specification version 2.5: Figure 90, Example High Speed Clock Enable, LP Mode. Figure 121 High Speed Transmission and Reception with HS-Idle Function.</i>
rx_skew_cal_hs_ck	1	Input	Calibration. Implementation-specific how the system responds to this condition. Refer to <i>MIPI D-PHY specification version 2.5: Figure 119. Skew Calibration, LP Mode.</i>
rx_alternate_cal_hs_ck	1	Input	
rx_error_cal_hs_ck	1	Input	
Data Lane High Speed Receive <lane>=0-7; <ppi_width>=16			
rx_srst_n_d<lane>	1	Input	Synchronous reset from MIPI RX D-PHY.
rx_word_clk_hs_d<lane>	1	Input	Refer to <i>MIPI D-PHY specification version 2.5: Figure 95 and 97, 101 High Speed Receive on Slave Side, 8/16-bit Bus, LP Mode.</i>
rx_data_width_hs_d<lane>	2	Output	
rx_data_hs_d<lane>	ppi_width	Input	
rx_valid_hs_d<lane>	ppi_width/8	Input	
rx_active_hs_d<lane>	1	Input	
rx_sync_hs_d<lane>	1	Input	
continued...			

continued...

Signal	Width	Direction	Description
rx_clk_active_hs_d<lane>	1	Input	Refer to <i>MIPI D-PHY specification version 2.5: Figure 90 Example High Speed Clock Enable, LP Mode. Figure 125 High Speed Transmission and Reception with HS-Idle Function.</i>
rx_skew_cal_hs_d<lane>	1	Input	Calibration. Implementation-specific how the system responds to this condition. Refer to <i>MIPI D-PHY specification version 2.5: Figure 119. Skew Calibration, LP Mode.</i>
rx_alterate_cal_hs_d<lane>	1	Input	
rx_error_cal_hs_d<lane>	1	Input	
Clock Lane Escape Mode Receive (LPDT = low-power data transmission, ULPS = Sleep Mode)			
rx_clk_esc_ck	1	Input	For LPDT. Required for USL control data communication. Refer to <i>MIPI D-PHY specification version 2.5: Figure 107 Example Low-Power Data Reception, LP Mode.</i>
rx_lpd_t_esc_ck	1	Input	
rx_data_esc_ck	8	Input	
rx_valid_esc_ck	1	Input	
rx_trigger_esc_ck	4	Input	For LPDT. The IP ignores any bit received after a Trigger Command (i.e., Reset Trigger) and before the Lines go to the Stop state. Refer to <i>MIPI D-PHY specification version 2.5: Figure 110-112.</i>
rx_ulps_esc_ck	1	Input	For ULPS and used with ulps_active_not_ck. Refer to <i>MIPI D-PHY specification version 2.5: Figure 116 Example Data Lane ULPS Entry and Exit, LP Mode.</i>
Data Lane Escape Mode Receive (LPDT = low-power data transmission, ULPS = Sleep Mode). <lane>=0-7			
rx_clk_esc_d<lane>	1	Input	For LPDT. Required for USL control data communication. Refer to <i>MIPI D-PHY specification version 2.5: Figure 107 Example Low-Power Data Reception, LP Mode.</i>
rx_lpd_t_esc_d<lane>	1	Input	
rx_data_esc_d<lane>	8	Input	
rx_valid_esc_d<lane>	1	Input	
rx_trigger_esc_d<lane>	4	Input	For LPDT. The IP ignores any bit received after a Trigger Command (i.e., reset trigger) and before the Lines go to the Stop state. Refer to <i>MIPI D-PHY specification version 2.5: Figure 110-112.</i>
rx_ulps_esc_d<lane>	1	Input	For ULPS and used with ulps_active_not_ck. Refer to <i>MIPI D-PHY specification version 2.5: Figure 116 Example Data Lane ULPS Entry and Exit, LP Mode.</i>
Clock Lane Control			
direction_ck	1	Input	Indicates the current direction of Lane. 0 = Transmit mode. 1 = Receive mode.
force_rx_mode_ck	1	Output	For Transmit: Refer to <i>MIPI D-PHY specification version 2.5: Figure 86. Master PHY Enable, LP Mode.</i>
force_tx_stop_mode_ck	1	Output	
stop_state_ck	1	Input	For Receive: Refer to <i>MIPI D-PHY specification version 2.5: Figure 88. Example Slave PHY Enable, LP Mode.</i>
enable_ck	1	Output	
alp_mode_ck	1	Output	Alternate Low Power Mode Selection.
continued...			

continued...

Signal	Width	Direction	Description
			0 = LP mode. 1 = ALP mode.
tx_ulps_clk_ck	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 114. Example Clock Lane ULPS Entry and Exit, LP Mode.</i>
rx_ulps_clk_not_ck	1	Input	
ulps_active_not_ck	1	Input	
tx_hsidle_clk_hs_ck	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 121. HS Transmission and Reception with HS-IDLE Function.</i>
tx_hsidle_clk_ready_hs_ck	1	Input	
Data Lane Control<lane>=0-7			
direction_d<lane>	1	Input	Indicates the current direction of Lane. 0 = Transmit mode. 1 = Receive mode.
force_rx_mode_d<lane>	1	Output	For Transmit: Refer to <i>MIPI D-PHY specification version 2.5: Figure 86. Master PHY Enable, LP Mode.</i> For Receive: Refer to <i>MIPI D-PHY specification version 2.5: Figure 88. Example Slave PHY Enable, LP Mode.</i>
force_tx_stop_mode_d<lane>	1	Output	
stop_state_d<lane>	1	Input	
enable_d<lane>	1	Output	
alp_mode_d<lane>	1	Output	Alternate Low Power Mode Selection. 0 = LP mode. 1 = ALP mode.
tx_ulps_clk_d<lane>	1	Output	Refer to <i>D-PHY specification version 2.5: Figure 114. Example Clock Lane ULPS Entry and Exit, LP Mode.</i>
rx_ulps_clk_not_d<lane>	1	Input	
ulps_active_not_d<lane>	1	Input	
tx_hsidle_clk_hs_d<lane>	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 121. HS Transmission and Reception with HS-IDLE Function.</i>
tx_hsidle_clk_ready_hs_d<lane>	1	Input	
Clock Lane Error Input			
i_err_sot_hs_ck	1	Input	Start-of-transmission error.
i_err_sot_sync_hs	1	Input	Start-of-transmission synchronization error.
i_err_esc_ck	1	Input	Escape entry error.
i_err_sync_ck	1	Input	Low-power data transmission synchronization error.
i_err_control_ck	1	Input	Control error.
i_err_contention_lp0_ck	1	Input	LP0 Contention error.
i_err_contention_lp1_ck	1	Input	LP1 Contention error.
Data Lane Error Input<lane>=0-7			
continued...			

Signal	Width	Direction	Description
i_err_sot_hs_d<lane>	1	Input	Start-of-transmission error.
i_err_sot_sync_d<lane>	1	Input	Start-of-transmission synchronization error.
i_err_esc_d<lane>	1	Input	Escape entry error.
i_err_sync_d<lane>	1	Input	Low-power data transmission synchronization error.
i_err_control_d<lane>	1	Input	Control error.
i_err_contention_lp0_d<lane>	1	Input	LP0 Contention error.
i_err_contention_lp1_d<lane>	1	Input	LP1 Contention error.
Clock Lane Error Passthrough Output			
o_err_sot_hs_ck	1	Input	Start-of-transmission error.
o_err_sot_sync_hs	1	Input	Start-of-transmission synchronization error.
o_err_esc_ck	1	Input	Escape entry error.
o_err_sync_ck	1	Input	Low-power data transmission synchronization error.
o_err_control_ck	1	Input	Control error.
o_err_contention_lp0_ck	1	Input	LP0 contention error.
o_err_contention_lp1_ck	1	Input	LP1 contention error.
Data Lane Error Passthrough Output<lane>=0-7			
o_err_sot_hs_d<lane>	1	Input	Start-of-transmission error.
o_err_sot_sync_d<lane>	1	Input	Start-of-transmission synchronization error.
o_err_esc_d<lane>	1	Input	Escape ntry error.
o_err_sync_d<lane>	1	Input	Low-power data transmission synchronization error.
o_err_control_d<lane>	1	Input	Control error.
o_err_contention_lp0_d<lane>	1	Input	LP0 Contention error.
o_err_contention_lp1_d<lane>	1	Input	LP1 Contention error.

Related Information

MIPI D-PHY specification

Describes the MIPI D-PHY specification version 2.5.

2.3.2. Avalon Memory-Mapped Interface Control Interface Signals

Table 9. Avalon Memory-Mapped Interface Control Interface Signals

Signal	Width	Direction	Description
control_address	10	Input	The Avalon memory-mapped interface agent port that provides access to internal control and status register. This interface is expected to operate at Nios V processor clock domain. Because messages can be large (more than 4 Bytes), the IP transfers the message in burst mode with full handshaking mechanism. Write transfers always have a wait time of 0 cycle. Read transfers have a wait time of 1 cycle. The addressing should be accessed as word addressing in the Platform Designer flow. For example, addressing of 4 in the Nios V software selects the address of 1 in the agent.
control_write	1	Input	
control_byteenable	4	Input	
control_writedata	32	Input	
control_read	1	Input	
control_readdata	32	Output	
control_readdatavalid	1	Output	
control_waitrequest	1	Output	Active high interrupt signal.
control_irq	1	Output	

2.3.3. Receiver AXI4-Stream Output Video Interface Signals

These MIPI CSI-2 IP signals follow the Altera Streaming Video protocol (full variant). The full variant of the protocol inserts Image Information Packets (IIPs) and End-of-Frame (EOF) packets on either side of each video frame sent. See the table below for information on the IIP values used for each data type. For more information about this protocol and rules for pixel formatting see the *Altera FPGA Streaming Video Protocol Specification*.

Table 10. IIP Image Property Values by Data Type

CSI-2 Data Type	Colorspace	Subsampling	Cositing	Bit Depth
RAW6	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	6-bits:0x05
RAW7	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	7-bits:0x06
RAW8	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	8-bits:0x07
RAW10	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	10-bits:0x09
RAW12	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	12-bits:0x0B
RAW14	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	14-bits:0x0D
continued...				

⁽²⁾ For Quartus Prime Pro Edition software version 26.1.1.

⁽³⁾ For Quartus Prime Pro Edition software version 26.1 and earlier.

CSI-2 Data Type	Colorspace	Subsampling	Cositing	Bit Depth
RAW16	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	16-bits:0x0F
RAW20	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	20-bits:0x13
RAW24	RAW:0x5	4:4:4:0x3⁽²⁾ - Unused:0x1⁽³⁾	Unused:0x0	24-bits:0x17
RGB444	RGB:0x0	4:4:4:0x3	Unused:0x0	4-bits:0x03
RGB555	RGB:0x0	4:4:4:0x3	Unused:0x0	5-bits:0x04
RGB565	RGB:565:0x6	4:4:4:0x3	Unused:0x0	6-bits:0x05
RGB666	RGB:0x0	4:4:4:0x3	Unused:0x0	6-bits:0x05
RGB888	RGB:0x0	4:4:4:0x3	Unused:0x0	8-bits:0x07
YUV420 8-bit	YUV:0x5	4:2:0:0x0	Center/center:0x3	8-bits:0x07
YUV420 8-bit (CSPS)	YUV:0x5	4:2:0:0x0	Center/left:0x2	8-bits:0x07
YUV420 8-bit (legacy)	YUV:0x5	4:2:0:0x0	Center/center:0x3	8-bits:0x07
YUV420 10-bit	YUV:0x5	4:2:0:0x0	Center/center:0x3	10-bits:0x09
YUV420 10-bit (CSPS)	YUV:0x5	4:2:0:0x0	Center/left:0x2	10-bits:0x09
YUV422 8-bit	YUV:0x5	4:2:2:0x2	Unused: 0x0	8-bits:0x07
YUV422 10-bit	YUV:0x5	4:2:2:0x2	Unused: 0x0	10-bits:0x09

Equation 1. Video Interface Bit Widths

$$P = \max \left(2, \text{ceil} \left(\frac{\text{bits_per_color_plane} \times \text{number_of_color_planes}}{8} \right) \right) \times \text{pixels_in_parallel} \times 8$$

$$Q = \text{ceil} \left(\frac{P}{8} \right)$$

Table 11. Examples for Video Interface Bit Widths

Data Type	bits_per_color_plane	number_of_color_planes
YUV422 8-bit	8	2
RGB565	6	3
RAW10	10	1

Table 12. Receiver AXI4-Stream Output Video Interface Signals

VC indicates virtual channel ID, between 0 and 3 according to IP configuration.

Signal	Width	Direction	Description
axi4s_vid_out_VC_tdata	P	Output	AXI4-Stream data out.
axi4s_vid_out_VC_tvalid	1	Output	AXI4-Stream data valid.

continued...

Signal	Width	Direction	Description
axi4s_vid_out_VC_tuser	Q	Output	Bit 0: AXI4-Stream start of video frame. 0 = Not start of field 1 = Start of field Bit 1: AXI4-Stream meta or data packet. 0 = Video packet 1 = Metapacket Bit Q-1:2: Unused
axi4s_vid_out_VC_tlast	1	Output	AXI4-Stream end of packet.
axi4s_vid_out_VC_tready	1	Input	AXI4-Stream data ready.

Related Information

[Streaming Video Protocol Specification](#)

2.3.4. Receiver AXI4-Stream Output Passthrough Interface

The MIPI CSI-2 IP uses this interface when in passthrough mode. It carries the MIPI CSI-2 packets recovered from the external MIPI input, as defined by the MIPI CSI-2 specification low-level protocol. This interface allows retransmission or custom processing of the received MIPI CSI-2 packets. For example in bridging applications where the IP combines multiple incoming CSI-2 streams into a single outgoing CSI-2 stream.

The IP combines MIPI CSI-2 packets from multiple lanes, if configured, but performs no further processing. The packets have packet header followed by payload and checksum for long packets.

The IP includes a single AXI4-Stream output interface, containing data from all virtual channels. According to your system requirements you can process this data as a combined stream or decode it further based on the virtual channel ID in the packet headers.

Table 13. Receiver AXI4-Stream Output Passthrough Interface Signals

$channel_width = PPI_data_width \times lane_count$

Signal	Width	Direction	Description
axi4s_mipi_out_tdata	<i>channel_width</i>	Output	Packet data
axi4s_mipi_out_tkeep	<i>channel_width</i> ÷ 8	Output	Packet tkeep
axi4s_mipi_out_tvalid	1	Output	Packet data_valid
axi4s_mipi_out_tuser	1	Output	Bit0: AXI4-Stream start of packet 0 = Not start of packet 1 = Start of packet
axi4s_mipi_out_tlast	1	Output	AXI4-Stream end of packet
axi4s_mipi_out_tready	1	Input	AXI4-Stream data ready

2.3.5. Transmitter PPI Signals

The interface between the Altera MIPI D-PHY IP and the MIPI CSI-2 IP uses the PPI defined in the MIPI D-PHY standards. This interface carries the outgoing data stream from the CSI-2 Transmitter that the D-PHY IP processes.

Table 14. Transmitter PPI Signals

Signal	Width	Direction	Description
Clock Lane High Speed Transmit<ppi_width>=16			
tx_word_clk_hs_ck	1	Input	Refer to MIPI D-PHY specification version 2.5: Figures 92-94 Example High-Speed Transmission from the Master Side: 8-/16-/32-Bit Bus Width.
tx_data_width_hs_ck	2	Output	
tx_data_hs_ck	ppi_width	Output	
tx_valid_hs_ck	ppi_width/8	Output	
tx_data_transfer_en_hs_ck	1	Output	
tx_request_hs_ck	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 90, Example High Speed Clock Enable, LP Mode. Figure 121 High Speed Transmission and Reception with HS-Idle Function.
tx_ready_hs_ck	1	Input	
tx_eq_active_hs_ck	1	Output	Refer to MIPI D-PHY specification version 2.5 Table 50.
tx_eq_level_hs_ck	1	Output	
tx_skew_cal_hs_ck	1	Output	Skew calibration. Implementation-specific how the system responds to this condition. Refer to MIPI D-PHY specification version 2.5: Figure 119. Skew Calibration, LP Mode.
tx_alternate_cal_hs_ck	1	Output	
Data Lane High Speed Transmit <lane>=0-7; <ppi_width>=16			
tx_word_clk_hs_d<lane>	1	Input	Refer to MIPI D-PHY specification version 2.5: Figures 92-94 Example High-Speed Transmission from the Master Side: 8-/16-/32-Bit Bus Width.
tx_data_width_hs_d<lane>	2	Output	
tx_data_hs_d<lane>	ppi_width	Output	
tx_valid_hs_d<lane>	ppi_width/8	Output	
tx_data_transfer_en_hs_d<lane>	1	Output	
tx_request_hs_d<lane>	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 90, Example High Speed Clock Enable, LP Mode. Figure 121 High Speed Transmission and Reception with HS-Idle Function.
tx_ready_hs_d<lane>	1	Input	
tx_eq_active_hs_d<lane>	1	Output	Refer to MIPI D-PHY specification version 2.5 Table 50.
tx_eq_level_hs_d<lane>	1	Output	
tx_skew_cal_hs_d<lane>	1	Output	Calibration. Implementation-specific how the system responds to this condition. Refer to MIPI D-PHY specification version 2.5: Figure 119. Skew Calibration, LP Mode.
tx_alternate_cal_hs_d<lane>	1	Output	
Clock Lane Escape Mode Transmit (LPDT= low-power data transmission, ULPS = Sleep Mode)			
continued...			

Signal	Width	Direction	Description
tx_clk_esc_ck	1	Input	Escape mode signals – Refer to <i>MIPI D-PHY specification version 2.5 Table 50</i> .
tx_request_esc_ck	1	Output	
tx_request_type_esc_ck	4	Output	
tx_lpdt_esc_ck	1	Output	
tx_ulps_exit_ck	1	Output	
tx_ulps_esc_ck	1	Output	
tx_trigger_esc_ck	4	Output	
tx_data_esc_ck	8	Output	
tx_valid_esc_ck	1	Output	
tx_ready_esc_ck	1	Input	
Data Lane Escape Mode Transmit (LPDT= low-power data transmission, ULPS = Sleep Mode). <lane>=0-7			
tx_clk_esc_d<lane>	1	Input	Escape mode signals – Refer to <i>MIPI D-PHY specification version 2.5 Table 50</i> .
tx_request_esc_d<lane>	1	Output	
tx_request_type_esc_d<lane>	4	Output	
tx_lpdt_esc_d<lane>	1	Output	
tx_ulps_exit_d<lane>	1	Output	
tx_ulps_esc_d<lane>	1	Output	
tx_trigger_esc_d<lane>	4	Output	
tx_data_esc_d<lane>	8	Output	
tx_valid_esc_d<lane>	1	Output	
tx_ready_esc_d<lane>	1	Input	
Clock Lane Control			
turn_request_ck	1	Output	Clock lane direction signals – Refer to <i>MIPI D-PHY specification version 2.5 Table 50</i> .
direction_ck	1	Input	
turn_disable_ck	1	Output	
force_rx_mode_ck	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 86. Master PHY Enable, LP Mode</i> .
force_tx_stop_mode_ck	1	Output	
stop_state_ck	1	Input	
enable_ck	1	Output	
alp_mode_ck	1	Output	Alternate Low Power Mode Selection - Refer to <i>MIPI D-PHY specification version 2.5 Table 50</i> .
tx_ulps_clk_ck	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 114. Example Clock Lane ULPS Entry and Exit, LP Mode</i> .
rx_ulps_clk_not_ck	1	Input	
ulps_active_not_ck	1	Input	
continued...			

Signal	Width	Direction	Description
tx_hsidle_clk_hs_ck	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 121. HS Transmission and Reception with HS-IDLE Function.</i>
tx_hsidle_clk_ready_hs_ck	1	Input	
Data Lane Control <lane>=0-7			
turn_request_d<lane>	1	Output	Data lane direction signals – Refer to <i>MIPI D-PHY specification version 2.5 Table 50.</i>
direction_d<lane>	1	Input	
turn_disable_d<lane>	1	Output	
force_rx_mode_d<lane>	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 86. Master PHY Enable, LP Mode.</i>
force_tx_stop_mode_d<lane>	1	Output	
stop_state_d<lane>	1	Input	
enable_d<lane>	1	Output	
alp_mode_d<lane>	1	Output	Alternate Low Power Mode Selection - Refer to <i>MIPI D-PHY specification version 2.5 Table 50.</i>
tx_ulps_clk_d<lane>	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 114. Example Clock Lane ULPS Entry and Exit, LP Mode.</i>
rx_ulps_clk_not_d<lane>	1	Input	
ulps_active_not_d<lane>	1	Input	
tx_hsidle_clk_hs_d<lane>	1	Output	Refer to <i>MIPI D-PHY specification version 2.5: Figure 121. HS Transmission and Reception with HS-IDLE Function.</i>
tx_hsidle_clk_ready_hs_d<lane>	1	Input	

2.3.6. Transmitter AXI4-Stream Input Video Interface Signals

These MIPI CSI-2 IP signals follow the Altera Streaming Video protocol (full variant). The full variant of the protocol must contain Image Information Packets (IIPs) before and End-of-Frame (EOF) packets after each video frame is sent. See the [IIP Image Property Values by Data Type](#) for information on the IIP values used for each data type. If the IIPs are not present or the values do not match the values shown in the table, it is not possible to determine the MIPI data type to be used. As a result, video is not transmitted. For more information on this protocol and rules for pixel formatting, refer to the *Altera FPGA Streaming Video Protocol Specification*.

Equation 2. Video Interface Bit Widths

$$P = \max \left(2, \text{ceil} \left(\frac{\text{bits per color plane} \times \text{number of color planes}}{8} \right) \right) \times \text{pixels in parallel} \times 8$$

$$Q = \text{ceil} \left(\frac{P}{8} \right)$$

Table 15. Examples for Video Interface Bit Widths]

Data Type	Bits per Color Plane	Number of color planes
RGB565	6	3
RAW10	10	1

Table 16. Transmitter AXI4-Stream Input Video Interface Signals

VC indicates virtual channel ID, between 0 and 3, according to IP configuration.

Signal	Width	Direction	Description
axi4s_vid_in_VC_tdata	<i>P</i>	Input	AXI4-Stream data out.
axi4s_vid_in_VC_tvalid	1	Input	AXI4-Stream data valid.
axi4s_vid_in_VC_tuser	<i>Q</i>	Input	Bit0: AXI4-Stream start of video frame. 0 = Not start of field 1 = Start of field Bit1: Meta or data packet. 0 = Video packet 1 = Metapacket Bits Q-1:2 = Unused
axi4s_vid_in_VC_tlast	1	Input	AXI4-Stream end of packet.
axi4s_vid_in_VC_tready	1	Output	AXI4-Stream data ready.

2.3.7. Transmitter AXI4-Stream Input Passthrough Interface

This interface carries MIPI CSI-2 packets in the same format as those produced by the CSI-2 receiver output passthrough interface. This interface allows retransmission of received MIPI CSI-2 packets, for example in bridging applications where the IP combines multiple incoming CSI-2 streams into a single outgoing CSI-2 stream.

The IP provides one AXI4-Stream interface per configured virtual channel. When you select multiple interfaces (**Number of video streaming interfaces** is greater than 1), the IP updates the virtual channel ID in the outgoing packet headers according to the AXI4-Stream channel where the packet resides and regenerates the ECC value in the header. When you select a single interface, the virtual channel ID in the packet headers passes through unaltered and ECC is not regenerated. You can use this configuration either in single-channel systems or where a merged multichannel stream is already formed.

When you select multiple interfaces, the IP multiplexes them at the input. The inputs have no buffering, so your system must consider the periods of backpressure on each channel while the IP processes other channels. For instance, include a FIFO buffer in the pipeline to each channel input if necessary in your system.

Table 17. Transmitter AXI4-Stream Input Passthrough Interface Signals

num_VCs indicates the number of virtual channels configured in the IP and *channel_width* = *PPI_data_width* × *lane_count*

Signal	Width	Direction	Description
axi4s_mipi_in_tdata	<i>num_VCs</i> × <i>channel_width</i>	Input	MIPI packet data. VC0 data are placed in the least-significant <i>channel_width</i>
continued...			

Signal	Width	Direction	Description
			bits, with the highest configured VC in the most significant bits.
axi4s_mipi_in_tkeep	$num_VCs \times (channel_width \div 8)$	Input	MIPI packet tkeep. VC0 signals are placed in the least-significant $channel_width \div 8$ bits, with the highest configured VC in the most significant bits.
axi4s_mipi_in_tvalid	num_VCs	Input	MIPI packet datavalid. VC0 signals are placed in the least-significant bit, with the highest configured VC in the most significant bit.
axi4s_mipi_in_tuser	num_VCs	Input	Bit0: AXI4-Stream start of packet 0 = Not start of packet 1 = Start of packet VC0 signals are placed in the least-significant bit, with the highest configured VC in the most significant bit.
axi4s_mipi_in_tlast	num_VCs	Input	AXI4-Stream end of packet. VC0 signals are placed in the least-significant bit, with the highest configured VC in the most significant bis.
axi4s_mipi_in_tready	num_VCs	Output	AXI4-Stream data ready. VC0 signals are placed in the least-significant bit, with the highest configured VC in the most significant bit.

3. MIPI CSI-2 IP Parameters

Table 18. IP Core Parameters

Parameter	Value	Description
Direction	RX, TX	Select the port direction.
Video Interface Mode	Video, Passthrough	Select Video to decode received CSI-2 to AXI4-Stream video pixel data or transmit CSI-2 from AXI4-Stream video pixel data", Select Passthrough to interface using MIPI CSI-2 packets and disable packet encode or decode.
Buffer depth	128, 256, 512, 1024, 2048, 4096, 8192, 16384	Select the FIFO buffer depth. For the receiver, the value is in pixels. For the transmitter, this value is in bytes. In both cases, a recommended minimum value contains 2 video lines. This value may need to be higher depending on the backpressure of the your system.
Scrambling (TX) or Descrambling (RX)	On or off	Turn on CSI-2 scrambling on transmitter or CSI-2 descrambling on receiver.
Number of lanes	1C & 1D, 1C & 2D, 1C & 4D, 1C & 8D	Select the number of D-PHY lanes. C indicates clock lane while D indicates Data lane count.
Error Correction Check (ECC) insertion (TX) or checking (RX)	On or off	Turn on for error correction code (ECC) processing. For the receiver, when Error correction check checking is off, the IP does not detect or correct any bit errors in the packet headers. For the transmitter, when Error correction check insertion is off, the IP does not calculate the header ECC code and it sends the value zero. If the transmitter has this option disabled the IP does not comply with the CSI-2 specification. Turning off this option allows selective reduction in IP resource utilization.
CRC insertion (TX) or checking (RX)	On or off	Cyclic redundancy check (CRC) processing. For the receiver, when CRC checking is off, the IP does not calculate long packet payload checksums so does not detect payload errors.. For the transmitter, when CRC insertion is off, the IP does not calculate long packet payload CRC, and it places the value zero in the field. For the transmitter, when CRC insertion is off, the IP does not comply with the CSI-2 specification. Turning off this parameter for either the transmitter or receiver gives a selective reduction in IP resource utilization.
Control and status registers	On or off	When control and status registers are off, the control and status register interface is not available, apart from the interface to the AXI2CV registers in the transmitter. Turning off this parameter allows selective reduction in IP resource utilization.
Separate clock for Control and Status Register interface (CSR)	On or off	This option allows control over the clock domain used on the Avalon memory-mapped interface CSR interface. When enabled, the Avalon memory-mapped interface signals operates on an independent domain, and <code>slave_clock</code> and <code>slave_reset</code> inputs ports is added to

continued...

Parameter	Value	Description
		the IP. When disabled, the Avalon memory-mapped interface is synchronous to the AXI4-S video clock domain.
PPI bus width per lane	16	Select the PHY Protocol Interface (PPI) bus width bits per lane.
MIPI D-PHY continuous clock	On or off	Transmitter: This controls the operation of the MIPI DPHY clock lane. When enabled, the clock lane is permanently active. When this option is disabled, the clock lane is activated throughout the active part of the video frame only and briefly enabled to allow timing packets to be sent during video blanking. The configuration of this must match the MIPI DPHY IP link connected to this IP. Receiver: When disabled, an additional clock input, <code>ls_clk</code>, is present on the interface to the IP. This is used to ensure that the received packets can be processed correctly. This clock must be the same frequency as the MIPI word clock, <code>rx_word_clk_hs_ck</code>. When enabled the word clock on the PPI is used directly.
Pixels in parallel	1, 2, 4	Select the number of pixels in parallel at the video streaming interface.
Number of video streaming interfaces	1, 2, 3, 4	Select the number of AXI-S video interfaces or CSI-2 virtual channels.
Support RAW6	On or off	Turn on to support the RAW6 video data type.
Support RAW7	On or off	Turn on to support the RAW7 video data type.
Support RAW8	On or off	Turn on to support the RAW8 video data type.
Support RAW10	On or off	Turn on to support the RAW10 video data type.
Support RAW12	On or off	Turn on to support the RAW12 video data type.
Support RAW14	On or off	Turn on to support the RAW14 video data type.
Support RAW16	On or off	Turn on to support the RAW16 video data type.
Support RAW20	On or off	Turn on to support the RAW20 video data type.
Support RAW24	On or off	Turn on to support the RAW24 video data type.
Support YUV420 8-bit	On or off	Turn on to support YUV420 8-bit video data type.
Support YUV420 8-bit (legacy)	On or off	Turn on to support YUV420 8-bit legacy video data type.
Support YUV420 10-bit	On or off	Turn on to support YUV420 10-bit video data type.
Support YUV422 8-bit	On or off	Turn on to support YUV422 8-bit video data type.
Support YUV422 10-bit	On or off	Turn on to support YUV422 10-bit video data type.
Support RGB444	On or off	Turn on to support the RGB444 video data type.
Support RGB555	On or off	Turn on to support the RGB555 video data type.
Support RGB565	On or off	Turn on to support the RGB565 video data type.
Support RGB666	On or off	Turn on to support the RGB666 video data type.
Support RGB888	On or off	Turn on to support the RGB888 video data type.



4. Designing with the MIPI CSI-2 IP

4.1. Generating the MIPI CSI-2 IP

1. Create or open an Quartus Prime project (.qpf) targeting a supported FPGA device family to include the IP variation.
2. In the IP catalog, navigate to **Interface Protocols > Audio & Video**.
3. Locate **MIPI CSI-2 IP**. You may type some or all the component's name in the IP Catalog search box to locate the IP. Double click on the IP. The **New IP Variation** window appears.
4. Specify a top-level name for this IP. Do not include spaces in IP names or paths. The parameter editor saves the IP variation settings in a file named <your_ip>.ip.
5. Click **OK**. The parameter editor appears.

Figure 1. Parameter Editor

The screenshot shows the 'Parameters' window for the 'MIPI CSI-2 IP' (intel_mipi_csi2). The window has a 'Details' button and a 'Generate Example Design...' button. The 'Design Example' tab is selected, showing the following options:

- General Options**
 - Direction: RX
 - Video Interface Mode: Video
 - Buffer depth: 16384
 - ☐ Scrambling (TX) or Descrambling (RX)
 - ☒ Error Correction Code (ECC) insertion (TX) or checking (RX)
 - ☒ CRC insertion (TX) or checking (RX)
 - ☒ Control and Status Registers (CSR)
 - ☐ Separate clock for Control and Status Register interface (CSR).
- DPHY Interface Options**
 - Number of lanes: 1C & 2D
 - PPI bus width per lane: 16 bits
 - ☒ MIPI D-PHY continuous clock
- Video Streaming Interface Options**
 - Number of video streaming interfaces: 1
 - Pixels in parallel: 2
- Video Packing/Unpacking Options**
 - It is recommended to use `axi4s_clk` with a ratio of at least 0.7 times `rx_word_clk_hs`.
 - ☐ Support YUV420 8-bit (legacy)
 - ☐ Support YUV420 8-bit
 - ☐ Support YUV420 10-bit
 - ☐ Support YUV422 8-bit

- Set your preferred parameter values in the parameter editor. The **Parameterization Messages** tab at the bottom displays any errors. To avoid unnecessary resource utilization, Altera advises that you only enable the features and data types required within the system in development.

7. Click **Generate HDL**. The **Generation** dialog box appears.
8. Specify output file generation options, and click **Generate**. Quartus Prime generates the synthesis and simulation files based on your selections.
9. Click **Finish** followed by **Yes** if prompted to add files representing the IP variation to your project.

4.2. MIPI CSI-2 IP Generation Output Quartus Prime Pro Edition

4.2.1. Files Generated for the MIPI CSI-2 IP and Platform Designer Systems

Figure 2. Files Generated for IP and Platform Designer Systems

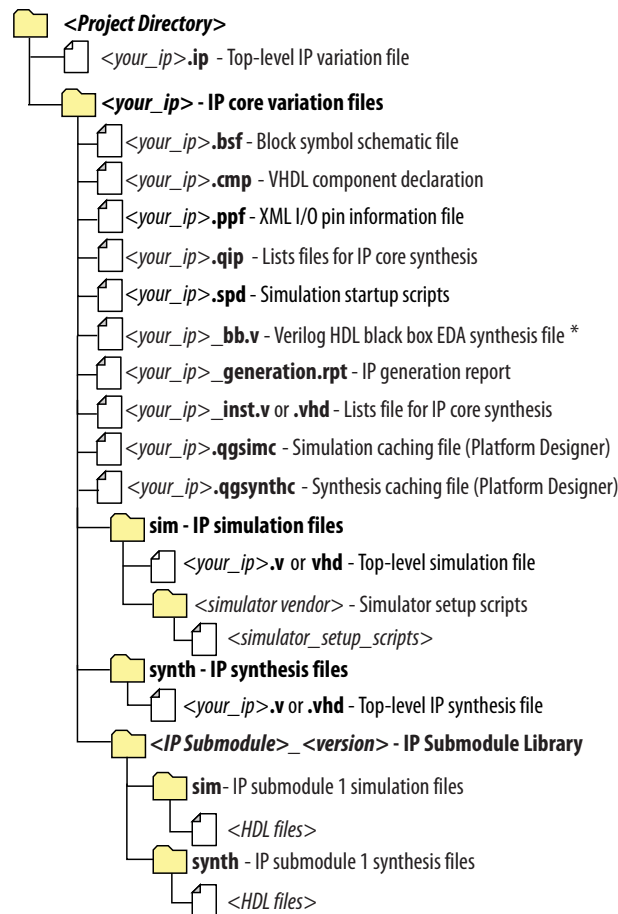


Table 19. IP and Platform Designer Simulation Files

File Name	Description
<my_ip>.cmp	The VHDL Component Declaration (.cmp) file is a text file that contains local generic and port definitions that you can use in VHDL design files.
<my_ip>_generation.rpt	IP or Platform Designer generation log file. A summary of the messages during IP generation.
<my_ip>.qgsimc	Simulation caching file that compares the .qsys and .ip files with the current parameterization of the Platform Designer system and IP. This comparison determines if Platform Designer can skip regeneration of the HDL.
<my_ip>.qgsynth	Synthesis caching file that compares the .qsys and .ip files with the current parameterization of the Platform Designer system and IP. This comparison determines if Platform Designer can skip regeneration of the HDL.
<my_ip>.qip	Contains all the required information about the IP component to integrate and compile the IP component in the Quartus Prime software.
<my_ip>.csv	Contains information about the upgrade status of the IP component.
<my_ip>.bsf	A Block Symbol File (.bsf) representation of the IP variation for use in Block Diagram Files (.bdf).
<my_ip>*.spd	Required input file for ip-make-simscript to generate simulation scripts for supported simulators. The .spd file contains a list of files generated for simulation, along with information about memories that you can initialize.
<my_ip>_bb.v	Use the Verilog black box (_bb.v) file as an empty module declaration for use as a black box.
<my_ip>_inst.v or _inst.vhd	HDL example instantiation template. Copy and paste the contents of this file into your HDL file to instantiate the IP variation.
<my_ip>.v <my_ip>.vhd	HDL files that instantiate each submodule or child IP for synthesis or simulation.
sim/mentor/	Contains a Questa* – Altera® FPGA Edition script msim_setup.tcl to set up and run a simulation.
sim/aldec/	Contains a Riviera-PRO script rivierapro_setup.tcl to setup and run a simulation.
sim/synopsys/vcsmx/	Contains a shell script vcsmx_setup.sh and synopsys_sim.setup file to set up and run a VCS MX® simulation.
sim/xcelium/	Contains a shell script xcelium_setup.sh and other setup files to set up and run a Xcelium simulation.
sim/common/	Contains a set of Tcl files, <simulator>_files.tcl, which provide all design related simulation information required by a corresponding simulation script. The Tcl file contains designs from current system-level hierarchy, and references to sub-systems and IP components.
<IP submodule>/	For each generated IP submodule directory, Platform Designer generates /synth and /sim sub-directories under IP-specific directories..

4.3. MIPI CSI-2 IP Systems Integration and Implementation

You must calculate the IP clock frequencies to ensure that the IP configuration is suitable to implement in hardware. You must assess the combination of clock rates, MIPI lane count, MIPI line rate, MIPI data type and AXI-S pixels-in-parallel to design a practical system.

4.3.1. D-PHY PPI Clock Frequencies

The MIPI DPHY IP generates high-speed word clocks on the PPI that the CSI-2 receiver and transmitter use to connect with the D-PHY.

Equation 3. D-PHY PPI Clock Frequencies

$$\text{rx_word_clk_hs_d0_freq} = \frac{\text{input_MIPI_lane_data_rate}}{\text{ppi_bus_width}}$$

$$\text{tx_word_clk_hs_d0_freq} = \frac{\text{output_MIPI_lane_data_rate}}{\text{ppi_bus_width}}$$

Table 20. MIPI D-PHY Clock Frequency of Fabric Interface

PPI Bus Width (Bits)	Lane Rate (Mbps)	rx or tx_word_clk_hs_d0 (MHz)
16	400	25
	1500	93.75
	2500	156.25

4.3.2. MIPI CSI-2 Receiver Clock Requirements

The MIPI CSI-2 receiver `axi4_clk` frequency must ensure that the output video data rate is the same or greater than the peak burst data rate on the external MIPI D-PHY interface. If not, data loss occurs.

Equation 4. Derivation of the CSI-2 Receiver `axi4_clk` Minimum Rate

$$\text{axi4s_clk} > \text{rx_word_clk_hs_d0} \times \frac{\text{lanes} \times \text{PPI bus width}}{\text{bits per pixel} \times \text{pixels in parallel}}$$

Altera recommends that you allow at least 10% buffer above the minimum clock rate calculated for this clock.

Table 21. MIPI CSI-2 Receiver AXI4-Stream Video Clock (`axi4s_clk`) Examples

Video Data Type	Lane Rate (Mbps)	PPI Bus Width (Bits)	rx_word_clk_hs_dX (MHz)	Lanes	Pixels in Parallel	Bits per Pixel	axi4s_clk (MHz)
RAW10	800	16	50	2	1	10	>160
RAW16	2500	16	156.25	4	4	16	>160
RGB888	1500	16	93.75	4	2	24	>125
YUV422 8-bit	1000	16	62.5	8	2	16	>250

4.3.3. MIPI CSI-2 Transmitter Clock Requirements

Select the MIPI CSI-2 transmitter `axi4s_clk` frequency such that the input video data rate achieved is the same or greater than the desired overall video data rate on the external MIPI D-PHY interface. If not, the data throughput does not meet requirements.

Equation 5. Derivation of the CSI-2 Transmitter `axi4_clk` Minimum Rate

$$\text{axi4s_clk} > \text{tx_word_clk_hs_d0} \times \frac{\text{lanes} \times \text{PPI bus width}}{\text{bits per pixel} \times \text{pixels in parallel}}$$

Altera recommends that you allow at least 10% buffer above the minimum clock rate calculated for this clock.

Table 22. MIPI CSI-2 Transmitter AXI4-Stream Video Clock (axi4s_clk) Examples

Video Data Type	Lane Rate (Mbps)	PPI Bus Width (Bits)	tx_word_clk_hs_dX (MHz)	Lanes	Pixels in Parallel	Bits per Pixel	axi4s_clk (MHz)
RAW10	800	16	50	2	1	10	>160
RGB888	1500	16	93.75	4	2	24	>125

Related Information

[Recommended Speed Grades for MIPI CSI-2 IP](#) on page 4

4.3.4. Required Supporting IP

You must connect the MIPI CSI-2 IP with the MIPI D-PHY IP. The MIPI D-PHY IP supports the external MIPI D-PHY signaling that carries the MIPI CSI-2 formatted data. For more information, refer to the *MIPI CSI-2 IP Design Example User Guide*.

Related Information

[MIPI CSI-2 IP Design Example User Guide](#)

4.3.5. Timing Configuration

The AXI2CV module video timing registers in the MIPI CSI-2 Transmitter IP must be configured via the AXI-S control interface at run-time to enable video to pass. The minimum set of registers that must be configured for virtual channel 0 are shown below. Additional controls are available, for further information see section [CSI-2 Transmitter Register Map](#).

0x160	VIDEO_INTF0_WIDTH[15:0]	VC0_WIDTH	Active pixels per line.
0x170	VIDEO_INTF0_HEIGHT_F0[15:0]	VC0_HEIGHT_F0	Active lines per frame in field 0.
0x180	VIDEO_INTF0_HEIGHT_F1[15:0]	VC0_HEIGHT_F1	Active lines per frame in field 1.
0x190	VIDEO_INTF0_LINE_BLANK[15:0]	VC0_LINE_BLANKING	Horizontal blanking pixels per line.
0x1A0	VIDEO_INTF0_FRAME_BLANK[15:0]	VC0_FRAME_BLANKING	Vertical blanking lines per frame.

The blanking register values must be non-zero. These values should be calculated based on the required line and frame interval, and clock period. The horizontal blanking period must consider the periods of transitions between the MIPI DPHY LP and HS states on each line, which can be significant especially when the horizontal resolution is relatively low. Typically, this should be at least 10% of the video line length. For shorter lines, this should be increased.

The MIPI CSI-2 Receiver IP does not require configuration at run-time.

4.3.6. Limitations and Known Issues

The MIPI CSI-2 IP supports CSI-2 v3.0 mandatory features when ECC, CRC, or scrambling is enabled. Resource-reduction options may violate specification, refer to the parameter notes.

Altera recommends that you read the information in the *MIPI D-PHY IP User Guide*, which describes the supported features in Agilex 3 and Agilex 5 FPGAs and the MIPI D-PHY IP. Each Agilex 5 device has specific limitations because of bonded pin availability. The B18A package provides a limited number of lanes for D-PHY interfaces, with available bonded pins between `BYTE4` to `BYTE7`. Conflicts may arise between the `RZQ` pin and data lane pins. For placement rules, refer to *MIPI D-PHY User Guide*.

In this release, the MIPI CSI-2 IP provides support for between one and four virtual channels. The IP supports 4-bit virtual channel IDs (VCIDs) using CSI-2 virtual channel extension (VCX) signaling, but only processes VCIDs 0 to 3. The IP does not process VCIDs 4 to 15 and drops any video data using these extended VCIDs. Packet headers must contain 6-bit ECC and 4-bit VCIDs. The receiver supports between one and four virtual channels in all data types except YUV420 formats, where one virtual channel only is supported.

Video line length and RX AXI-S line blanking must be an integer multiple of the CSI-2 data type packing multiple. For example, multiples of four pixels for RAW10, where four 10-bit pixels are packed into five bytes. The MIPI CSI-2 specification defines the packet data size constraints for each data type.

The IP supports interlaced video only when you insert MIPI line start and line end packets into the stream according to the CSI-2 standard.

When operating with relatively low DPHY line rates, e.g. below 400 Mbps, the MIPI CSI-2 receiver may not produce a video output if the MIPI source does not drive a continuous clock on the clock lane. The reason is the source provides insufficient clock pulses in the HS-TRAIL period following HS data transmission to process the received packet. This can be resolved in a number of ways:

- Increase the `TCLK-POST` or `THS-TRAIL` values on your MIPI CSI-2 source.
- Enable continuous clock mode on the MIPI input source.
- Disable the MIPI DPHY continuous option on the MIPI CSI-2 RX IP. When this option is disabled, an additional clock and reset input (`ls_clk` and `ls_rst`) are present on the MIPI CSI-2 RX IP. These should be connected to a continuous clock generated within the FPGA at the same frequency as the word clock from the MIPI DPHY RX IP: `rx_word_clk_hs_ck`. Incoming packets are transferred onto this clock for processing to minimize the dependency on the received MIPI clock.

When upgrading MIPI CSI-2 Receiver IP originally generated in Quartus Prime Pro Edition version 24.2, you may need to delete and recreate the MIPI CSI-2 IP because of a parameter problem in that version.

Related Information

[MIPI D-PHY IP User Guide: Agilex 5 FPGAs](#)

4.3.7. Troubleshooting

In most instances the MIPI CSI-2 IP does not require active monitoring. If the video transfer is unreliable, you can try various steps to isolate the problem. Initially confirm that the build-time configuration of the MIPI D-PHY and MIPI CSI-2 IPs match those configured in your external MIPI source or sink. Also check that the Quartus Prime project pin assignments correlate with the PCB layout.

The CSI-2 IP does not output video until 5 consecutive stable frames are detected, which reduces the likelihood the IP produces corrupted video data. If the video input is unstable, you see no video output.

1. For the receiver:
 - a. Read the error IRQ status registers in the CSI-2 IP to determine whether the IP detected any errors on the incoming video stream.
 - b. Check the CSI-2 IP VCx_VIDEO_DATA_TYPE registers record the most recent video data type received on each virtual channel. The registers show that video packets arriving at the CSI-2 input are of the expected data type and that the virtual channel ID in the video packets is correct. Registers reading zero, indicate the IP received no valid video packets on this channel.
 - c. Check the VIDEO_INTFx_WIDTH/HEIGHT/COLOR_PATTERN registers in the CSI-2 IP record the most recent video data characteristics received from the MIPI source. If these registers fluctuate, check for a signal integrity issue on the MIPI transmission line to the FPGA. If these registers are incorrect but constant, check the configuration of the MIPI source.
 - d. Check the VIDEO_INTFx_STABLE register indicates that video data on each channel remains active and has constant dimensions. Dimensions available in the registers above but with no stable flag indicate an unreliable input to the FPGA.
 - e. Ensure that the AXI-S video streaming output from the CSI-2 IP can flow adequately, as insufficient AXI throughput may cause buffer overflow, resulting in data loss or corrupted video data
 - Monitor data flow with the Signal Tap logic analyzer and ensure sufficient bandwidth in general, by comparing the video line rate on the PPI between the D-PHY and CSI-2 IPs with that on the output of the CSI-2 IP. A lower CSI-2 line rate indicates insufficient flow.
 - To confirm, tie the output video AXI-S TREADY signals to logic '1' and observe to see if the problem is resolved. If the problem is resolved, the system architecture does not support the required bandwidth. Select a higher number of pixels-in-parallel on the CSI-2 video output to increase the bandwidth (and update the downstream logic). Alternatively, change the downstream logic to reduce cycles where TREADY is not asserted to improve throughput.
2. For the transmitter:
 - a. Ensure that the source to the video input is producing valid Altera Video Streaming Protocol-compliant data, including the Image Information Packets (IIPs). If the source does not produce IIPs then the Altera VVP Protocol Converter IP can be used to insert these packets.

- b. Confirm that the IIPs contain the correct image properties for the MIPI CSI-2 data type required as these must match those shown in the [IIP Image Property Values by Data Type](#) table. The Altera VVP Protocol Converter IP can be used to modify the values in existing IIPs before the MIPI CSI-2 transmitter IP video input.
- c. Check that the IP has been configured to enable support for the MIPI CSI-2 data type that matches the IIP.
- d. Confirm that the video parameters in the VIDEO_INTFx_WIDTH/HEIGHT/COLOR_PATTERN and VCx_LINE-BLANK/FRAME_BLANK registers in the CSI-2 IP is configured correctly.
- e. Check that the VCx_MODE_MATCH registers in the CSI-2 IP indicate that the incoming video dimensions match the configured values. If the mode match bits for active channels are not set, check if the size of the incoming video matches the registers above.
- f. Ensure that video flow into the IP is sufficient to provide enough video data to drive the output at the rate configured. If video data is not available to send at the configured line or frame interval, the IP skips video packets.
- g. Check error status in the MIPI sink in your system. Errors may indicate signal integrity issues.

5. MIPI CSI-2 IP Block Descriptions

Related Information

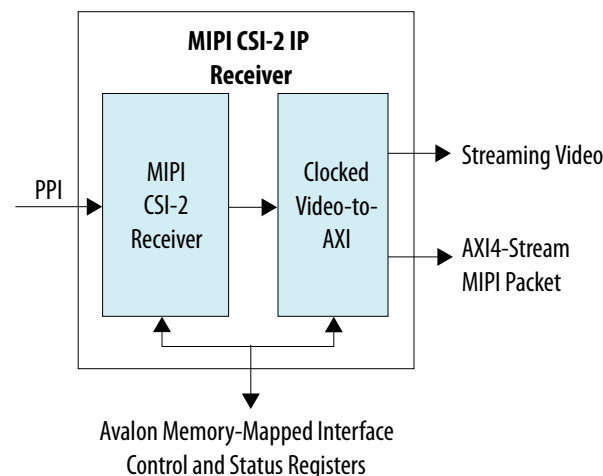
MIPI D-PHY specification

Describes the MIPI D-PHY specification version 2.5.

5.1. MIPI CSI-2 Receiver

The MIPI CSI-2 receiver consists of multiple layers defined in the MIPI CSI-2 specification version 3.0, such as the lane management layer, low level protocol and byte to pixel conversion. The MIPI CSI-2 receiver receives 16-bit data per lane, with support for up to eight lanes, from the MIPI D-PHY IP through the standard PPI.

Figure 3. MIPI CSI-2 IP Receiver Block Diagram



In video mode, the low-level protocol module processes the byte data it receives on the PPI to extract the real video information. The extracted and formatted video is available on the native clocked video interface.

In passthrough mode, the low-level protocol byte stream is available directly on an AXI4-Stream interface for retransmission or custom processing.

The Clocked Video to AXI converter primarily converts the clocked video data to AXI4-Stream video data using the Altera FPGA Streaming Video Protocol. The AXI4-Stream Altera FPGA streaming video packets always follow the pattern of one (and only one) image information metapacket followed by video packets and one (and only one) end-of-field metapacket.

Each component has an Avalon memory-mapped register interface for control and status register access and is given an address space. Bits [9:8] of the offset addresses from the system base address are 0x0 for MIPI CSI-2 RX component and 0x1 for the Clocked Video to AXI converter.

Table 23. Top Level Interfaces

Interface	Description
AXI4-Stream Altera FPGA Streaming Video (Full Variant)	Carries pure video packet in Altera FPGA streaming video format. Color and byte remapped and ready for processing. Accompanied by image information packet and end of field packets. Available only when you select Video mode.
AXI4-Stream MIPI Packet	AXI4-Stream interface carrying recovered MIPI CSI-2 packets. Available only when you select Passthrough mode.
PPI	PHY Protocol Interface. Refer to <i>MIPI D-PHY specification version 2.5 Annex A</i> .
Avalon memory-mapped interface control and status registers	Optional control and status register.

Figure 4. MIPI CSI-2 IP Receiver (Video Mode)

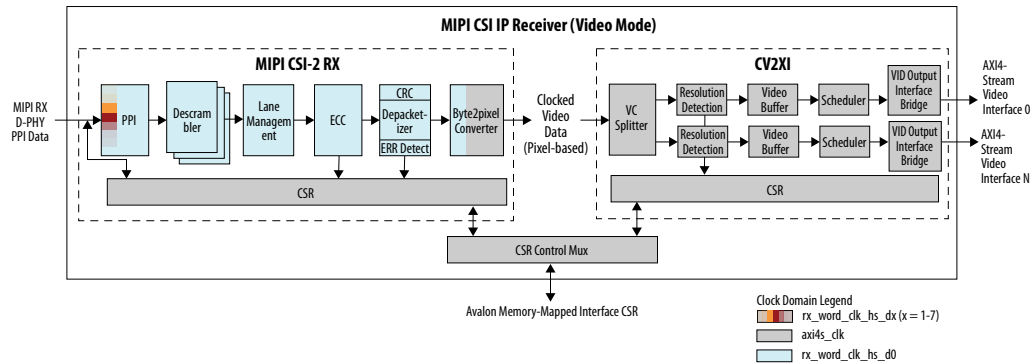


Figure 5. MIPI CSI-2 IP Receiver (Passthrough Mode)

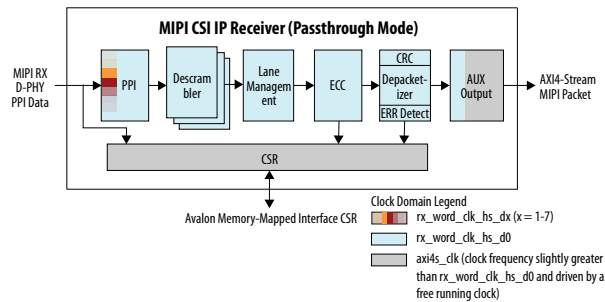


Table 24. MIPI CSI-2 IP RX Functional Blocks

Functional Block	Description
RX PPI	- Synchronizes all D-PHY data lanes to a single clock domain (<code>rx_word_clk_hs_d0</code>). When the MIPI continuous clock input parameter is disabled, the received data is then transferred onto <code>ls_clk</code>. - Processes lanes de-skew if more than one data lane is in use. - For easier packet processing downstream, it generates start-of-packet (SOP) and end-of-packet (EOP) markers for each received packet.
Descrambling	- Recovers the original CSI-2 data stream by reversing the CSI-2 LFSR scrambling on the video long packets. - Present when scrambling/descrambling is enabled when configuring the IP.
Lane Management	Performs lane merging function which collects incoming bytes from N number of D-PHY lanes and consolidates them into complete packets to pass on to subsequent low level protocol layers.
Error Correction Code (ECC) checking	- Generates a new ECC for the received packet header or short packet, computes the syndrome using the new ECC and the received ECC, decodes the syndrome to find if a single bit error has occurred, and if so, corrects it. - If more than one bit error occurs, an uncorrectable error occurs, and an error flag is asserted in the control and status register).
Depacketizer	- Extracts Packet Header, Packet Footer, HS-trail and all short packets. - Generates important markers such as start of frame (SOF), interlaced, even field or line, data type, and Virtual Channel identifier for downstream modules.
CRC and Error checking	- Calculates the checksum of the payload data of every long packet to detect possible errors in transmission. - The IP compares the calculated checksum against the received 16-bit cyclic redundancy check (CRC) checksum from the packet footer. The IP flags an error for a mismatch. - The error detect submodule also detects frame synchronization error when a frame end (FE) is not paired with a frame start (FS) on the same virtual channel.
Byte-to-Pixel Converter	- The video payload data from the depacketizer is unpacked based on the video data type information. The pixel output is based on the requested number of pixels in parallel. - An asynchronous FIFO buffer transfers the byte-based video payload data from <code>rx_word_clk_hs_d0</code> clock domain to <code>axi4s_clk</code> clock domain. Ensure that you select <code>axi4s_clk</code> so that the output bandwidth is greater than the link bandwidth rather than the underlying video data rate, as the bandwidth peaks at the MIPI line rate.
continued...	

Functional Block	Description
Clocked Video to AXI converter	- Converts clocked video data from the MIPI CSI-2 RX submodule to AXI4-Stream video data. - Resolution detection submodule retrieves the video attributes such as number of pixels per line, number of video lines per field, number of bits per symbol, color space, and interlaced or progressive at the control and status registers. - Stable bit is generated once video line with consistent width and height is detected. Until the input is stable, the IP produces no output video. This takes five video frames. - A video line buffer is present to accommodate temporary throttling on output video interface. Configure the buffer depth in the GUI to hold at least two video lines of data. - The scheduler ensures that the outgoing AXI4-Stream video data adheres to Altera FPGA Streaming Video (Full variant) video protocol.
Aux Output	- Buffers the MIPI packets and transfers to axi4s_clk. - Present when Passthrough mode is on when generating the IP only.
Control and status Register	- MIPI CSI-2 receiver and Clocked Video to AXI converter each have their own registers. - A control MUX resides at the higher layer to multiplex the control operation between both control and status registers. - The MSBs of the address select between accessing the MIPI CSI-2 receiver and Clocked Video to AXI converter registers. - When the separate slave clock option is enabled, a clock-crossing bridge is added between the external port and the internal logic. This allows external accesses on an independent clock domain.

Related Information

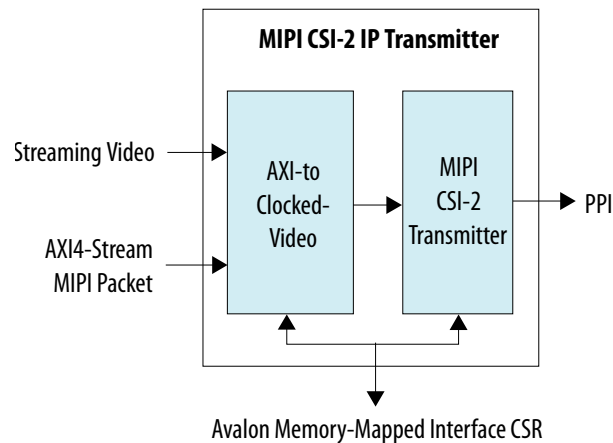
MIPI D-PHY specification

Describes the MIPI D-PHY specification version 2.5.

5.2. MIPI CSI-2 Transmitter

The MIPI CSI-2 transmitter consists of multiple layers defined in the MIPI CSI-2 specification version 3.0, such as the pixel to byte conversion, low level protocol and lane management layer.

Figure 6. MIPI CSI-2 IP Transmitter Block Diagram



The AXI to clocked video converter receives video in the Altera FPGA Streaming Video Protocol via AXI4-Stream and converts it to clocked video format.

In video mode, the MIPI CSI-2 transmitter converts the video data into the selected CSI-2 video data type and wraps it in CSI-2 packets according to the defined low-level protocol, adding synchronization packets as required. The IP calculates long packet checksums and header ECC values to allow it to detect or correct transmission errors.

In passthrough mode, it receives MIPI CSI-2 packets on an AXI4-Stream input interface from your logic, and bypasses the video mode processing.

The IP distributes CSI-2 packets across the MIPI data lanes and to the MIPI D-PHY IP via the standard PPI.

Each component has an Avalon memory-mapped interface register interface for control and status register access and is given an address space. Bits [9:8] of the offset addresses from the system base address are 0x0 for the MIPI CSI-2 transmitter and 0x1 for the AXI to clocked video converter.

Table 25. Top Level Interfaces

Interface	Description
AXI4-Stream Altera FPGA Streaming Video (Full Variant) input	Carries pure video packet in Altera FPGA streaming video format. Includes image information packet and end of field packets. Available only when you select Video mode.
AXI4_Stream MIPI Packet	AXI4-Stream input interface carrying MIPI CSI-2 packets. Available only when you select Passthrough mode.
TX PPI	PHY-Protocol Interface. Refer to <i>MIPI D-PHY specification version 2.5 Annex A</i> .
Avalon memory-mapped interface control and status register	Control and status register. Connection to CSI-2 transmitter is optional.

Figure 7. MIPI CSI-2 IP Transmitter (Video Mode)

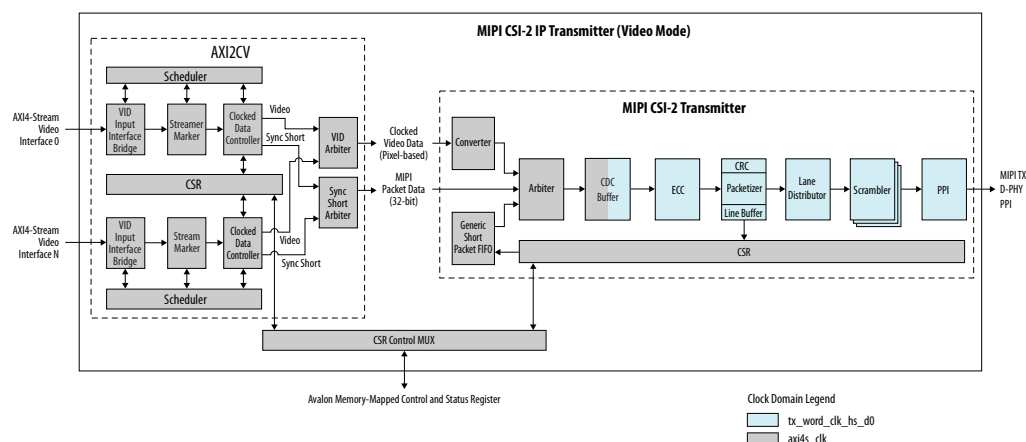


Figure 8. MIPI CSI-2 IP Transmitter (Passthrough Mode)

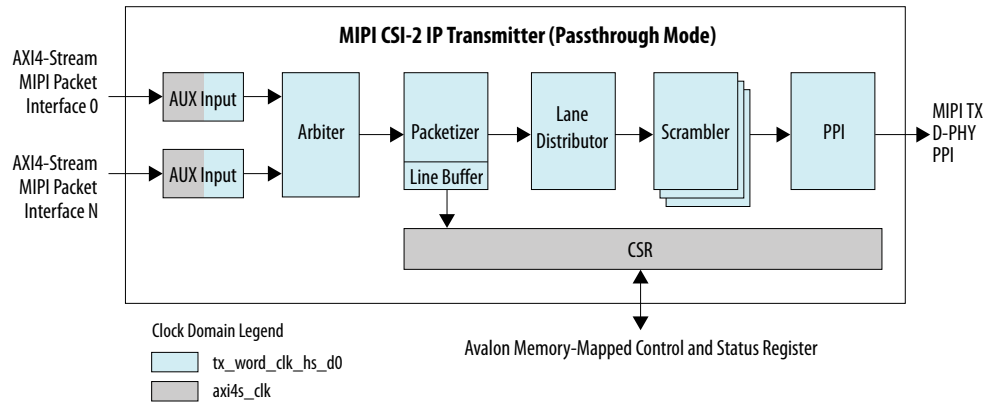


Table 26. MIPI CSI-2 Transmitter Functional Blocks

Functional Block	Description
AXI to clocked video converter	- Receives AXI4-Stream Altera Video Streaming Protocol (full variant) data and converts to clocked video data for the MIPI CSI-2 transmitter. - Generates synchronization short packets for insertion into the CSI-2 data stream around the video packets - Supports multiple channels – arbitrates between input channels in round-robin order. The IP uses the channel number to set the virtual channel ID in the MIPI CSI-2 data stream.
Pixel-to-byte converter	- Converts the video data from pixel format to the selected MIPI CSI-2 data type as defined in the CSI-2 specification. - Packs pixel data into bus with width that ensures that the MIPI link is fully occupied.
Packet arbiter	Multiplexes synchronization short packets, video long packets and generic short packets into a single stream.
Error Correction Code (ECC)	Calculates the ECC for each packet header.
Packetizer	- Applies long packet header. - Long packet payload CRC is calculated and appended after payload.
Lane distributor	Distributes the bytes from the incoming packet stream onto the configured number of MIPI lanes according to the CSI-2 specification.
Scrambler	When configured this applies the CSI-2 linear feedback shift register (LFSR) scrambling function to long packet payloads to avoid any long periods of fixed signaling when the video data is constant.
PPI	Performs handshaking with D-PHY IP to transmit CSI-2 packets. When the MIPI continuous clock option is enabled, the clock lane is held permanently active in HS mode. When disabled, the clock lane is active throughout the active video region of each frame and disabled during blanking except when sending short packets.
Control and status register	- MIPI CSI-2 transmitter and AXI to clocked video converter submodules each have their own CSRs. - A control MUX resides at the higher layer to multiplex the control operation between both control and status registers. - The MSBs of the address select between accessing the MIPI CSI-2 transmitter and AXI to clocked video converter control and status registers. - When the separate slave clock option is enabled, a clock-crossing bridge is added between the external port and the internal logic. This allows external accesses on an independent clock domain.

Related Information

[MIPI D-PHY specification](#)

Describes the MIPI D-PHY specification version 2.5.

6. Registers

6.1. CSI-2 Receiver Register Map

Base address (`control_address[9:8]`) 0x0 accesses the MIPI CSI-2 receiver control and status registers. Base 0x1 accesses the CV2AXI registers. When you turn off **Enable Control and Status Registers**, you have no access to any of the registers in this section.

Table 27. Register Access Codes

Code	Description
RW	Read and write.
RO	Read only.
RW1C	Read, write and clear. Your application writes 1 to the register bit(s) to invoke a defined instruction. The IP clears the bit(s) on running instructions.

Table 28. CSI-2 Receiver IP Registers

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
0x000	CORE_STATUS	RO	0x0	31:3	RESERVED	Reserved.
			LANES -1	2:0	MAXIMUM_LANES	Number of D-PHY lanes configured during IP generation minus 1. For example, 4 lanes = 3.
0x001 – 0x003	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x004	GENERIC_SHORT_PACKET	RO	0x0	31:26	RESERVED	Reserved.
				25:8	DATA	16 bits payload data of the generic short packet.
				7:6	VIRTUAL_CHANNEL	Virtual channel identifier.
				5:0	DATA_TYPE	Generic short packet code.
0x005 – 0x00F	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x010	VC0_IMAGE_INFO	RO	0x0	31:6	RESERVED	Reserved.
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
				5:0	VC0_VIDEO_DATA_TYPE	Data type of current packet (0x18 and above) received on VC0.
0x011	VC1_IMAGE_INFO	RO	0x0	31:6	RESERVED	Reserved.
				5:0	VC1_VIDEO_DATA_TYPE	Data type of current packet (0x18 and above) received on VC1.
0x012	VC2_IMAGE_INFO	RO	0x0	31:6	RESERVED	Reserved.
				5:0	VC2_VIDEO_DATA_TYPE	Data type of current packet (0x18 and above) received on VC2.
0x013	VC3_IMAGE_INFO	RO	0x0	31:6	RESERVED	Reserved.
				5:0	VC3_VIDEO_DATA_TYPE	Data type of current packet (0x18 and above) received on VC3.
0x014 – 0x01F	VC<ch>_IMAGE_INFO	RO	0x0	31:6	RESERVED	Reserved.
				5:0	VC<ch>_VIDEO_DATA_TYPE	Data type of current packet (0x18 and above) received on VC<ch>. <ch>: 4-15
0x020 – 0x050	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x051	DATA_TYPE_ERROR_IRQ_STAT US	RW1C	0x0	31:16	RESERVED	Reserved.
				15:1	VC<ch>_DT_ERROR	Interrupt status registers for data type error indicator on VC1 through VC15.
				0	VC0_DT_ERROR	Interrupt status register. Asserts when a packet header is decoded with an unrecognized or not implemented ID on VC0.
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
0x052	FRAME_SYNC_ERROR_IRQ_STATUS	RW1C	0x0	31:16	RESERVED	Reserved.
				15:1	VC<ch>_FRAME_SYNC_ERR	Interrupt status registers for frame sync error indicator on VC1 through VC15.
				0	VC0_FRAME_SYNC_ERR	Interrupt status register. Asserts when an FE is not paired with a FS on VC0.
0x053	FRAME_DATA_ERROR_IRQ_STATUS	RW1C	0x0	31:16	RESERVED	Reserved.
				15:1	VC<ch>_FRAME_DATA_ERR	Interrupt status registers for frame CRC error indicator on VC1 through VC15.
				0	VC0_FRAME_DATA_ERR	Interrupt status register. Asserts when the data payload received between FS and FE contains CRC errors on VC0.
0x054	ECC_1BIT_ERROR_IRQ_STATUS	RW1C	0x0	31:16	RESERVED	Reserved.
				15:1	VC<ch>_ECC_1BIT_ERR	Interrupt status registers for ECC single bit error indicator on VC1 through VC15.
				0	VC0_ECC_1BIT_ERR	Interrupt status register. Asserts when the IP calculates an ECC syndrome and it detects and corrects a single bit error in the packet header on VC0.
0x055	ECC_2BIT_ERROR_IRQ_STATUS	RW1C	0x0	31:16	RESERVED	Reserved.
				15:1	VC<ch>_ECC_2BIT_ERR	Interrupt status registers asserted when the IP detects an uncorrectable
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
						2-bit ECC error on VC1 through 15.
				0	VC0_ECC_2BIT_ERR	Interrupt status register. Asserts when the IP calculates an ECC syndrome and detects an uncorrectable 2-bit error in the received packet header on VC0.
0x056 – 0x06F	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x070	CSI2_RX_INTERRUPT_MASK_ENABLE	RW	0x0	31:6	RESERVED	Reserved.
				5	IRQ_EN_ECC_2BIT_ERR	Interrupt enable registers allow you to selectively generate an interrupt at the output port for each error bit in the registers 0x51 through 0x55. Set to 0 to disable the interrupt.
				4	IRQ_EN_ECC_1BIT_ERR	
				3	IRQ_EN_FRAME_DATA_ERR	
				2	IRQ_EN_FRAME_SYNC_ERR	
				1	IRQ_EN_DATA_TYPE_ERR	
				0	RESERVED	Reserved.
0x071 – 0x14F	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x150	VIDEO_INTF0_STATUS	RO	0x0	31:2	RESERVED	Reserved.
				1	VC0_INTERLACED	Interlaced bit of video streaming interface 0. When asserted, the input video stream is interlaced. Otherwise, the input video stream is progressive.
				0	RESERVED	Reserved.
0x151	VIDEO_INTF1_STATUS	RO	0x0	31:2	RESERVED	Reserved.
				1	VC1_INTERLACED	Interlaced bit of video streaming interface 1. When

continued...

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
						asserted, the input video stream is interlaced. Otherwise, the input video stream is progressive.
				0	RESERVED	Reserved.
0x152	VIDEO_INTF2_STATUS	RO	0x0	31:2	RESERVED	Reserved.
				1	VC2_INTERLACED	Interlaced bit of video streaming interface 2. When asserted, the input video stream is interlaced. Otherwise, the input video stream is progressive.
				0	RESERVED	Reserved.
0x153	VIDEO_INTF3_STATUS	RO	0x0	31:2	RESERVED	Reserved.
				1	VC3_INTERLACED	Interlaced bit of video streaming interface 3. When asserted, the input video stream is interlaced. Otherwise, the input video stream is progressive.
				0	RESERVED	Reserved
0x154 – 0x15F	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x160	VIDEO_INTF0_WIDTH	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC0_WIDTH	Specifies the active video width (in pixel unit) for video streaming interface 0.
0x161	VIDEO_INTF1_WIDTH	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC1_WIDTH	Specifies the active video width (in pixel unit) for video streaming interface 1.
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
0x162	VIDEO_INTF2_WIDTH	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC2_WIDTH	Specifies the active video width (in pixel unit) for video streaming interface 2.
0x163	VIDEO_INTF3_WIDTH	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC3_WIDTH	Specifies the active video width (in pixel unit) for video streaming interface 3.
0x164 – 0x16F	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x170	VIDEO_INTF0_HEIGHT_F0	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC0_HEIGHT_F0	Specifies the active video height (in line unit) of progressive video or interlaced video field 0 for video streaming interface 0.
0x171	VIDEO_INTF1_HEIGHT_F0	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC1_HEIGHT_F0	Specifies the active video height (in line unit) of progressive video or interlaced video field 0 for video streaming interface 1.
0x172	VIDEO_INTF2_HEIGHT_F0	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC2_HEIGHT_F0	Specifies the active video height (in line unit) of progressive video or interlaced video field 0 for video streaming interface 2.
0x173	VIDEO_INTF3_HEIGHT_F0	RO	0x0	31:16	RESERVED	Reserved.
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
				15:0	VC3_HEIGHT F0	Specifies the active video height (in line unit) of progressive video or interlaced video field 0 for video streaming interface 3.
0x174 – 0x17F	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x180	VIDEO_INTF0_HEIGHT_F1	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC0_HEIGHT_F1	Specifies the active video height (in line unit) for interlaced video field 1 for video streaming interface 0.
0x181	VIDEO_INTF1_HEIGHT_F1	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC1_HEIGHT F1	Specifies the active video height (in line unit) for interlaced video field 1 for video streaming interface 1.
0x182	VIDEO_INTF2_HEIGHT_F1	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC2_HEIGHT F1	Specifies the active video height (in line unit) for interlaced video field 1 for video streaming interface 2.
0x183	VIDEO_INTF3_HEIGHT_F1	RO	0x0	31:16	RESERVED	Reserved.
				15:0	VC3_HEIGHT F1	Specifies the active video height (in line unit) for interlaced video field 1 for video streaming interface 3.
0x184 – 0x18F	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x190	VIDEO_INTF0_COLOR_PATTERN	RO	0x0	31:13	RESERVED	Reserved
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
			0xF	12:9	VC0_COLOR_SPACE	Specifies the color space for video streaming interface 0. 0: RGB 1-4: Reserved 5: RAW 6: RGB565 7: YUV 8: YUV legacy 9-15: Reserved
			0x0	8:7	VC0_COSITE	Specifies the co-siting of the chroma samples for video streaming interface 0. 0: Reserved. 1: Top and center (ie. YUV420 8/10-bit Chroma Shifted Pixel Sampling). 2: Reserved. 3: Center and center (YUV420 8/10-bit).
			0x1	6:5	VC0_SUBSAMPLING	Specifies the chroma sampling for video streaming interface 0. 0: 420. 1: Reserved. 2: 422. 3: 444.
			0x0	4:0	VC0_BIT_WIDTH_-1	Specifies the bit width – 1 for video streaming interface 0. I.e., value of 5'd23 indicates bit width of 24.
0x191	VIDEO_INTF1_COLOR_PATTER N	RO	0x0	31:13	RESERVED	Reserved.
			0xF	12:9	VC1_COLOR SPACE	Specifies the color space for video streaming interface 1. 0: RGB 1-4: Reserved
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
						5: RAW 6: RGB565 7: YUV 8: YUV legacy 9-15: Reserved
			0x0	8:7	VC1_COSITE	Specifies the co-siting of the chroma samples for video streaming interface 1. 0: Reserved 1: Top and center (ie. YUV4208/10-bit Chroma Shifted Pixel Sampling) 2: Reserved 3: Center and center (YUV420 8/10-bit)
			0x1	6:5	VC1_SUBSAMPLING	Specifies the chroma sampling for video streaming interface 1. 0: 420 1: Reserved 2: 422 3: 444
			0x0	4:0	VC1_BIT WIDTH MINUS 1	Specifies the bit width – 1 for video streaming interface 1. I.e., value of 5'd23 indicates bit width of 24.
0x192	VIDEO_INTF2_COLOR_PATTER N	RO	0x0	31:13	RESERVED	Reserved.
			0xF	12:9	VC2_COLOR SPACE	Specifies the color space for video streaming interface 2. 0: RGB 1-4: Reserved 5: RAW 6: RGB565 7: YUV 8: YUV legacy 9-15: Reserved
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
			0x0	8:7	VC2_COSITE	Specifies the co-siting of the chroma samples for video streaming interface 2. 0: Reserved 1: Top and center (ie. YUV4208/10-bit Chroma Shifted Pixel Sampling) 2: Reserved 3: Center and center (YUV420 8/10-bit)
			0x1	6:5	VC2_SUBSAMPLING	Specifies the chroma sampling for video streaming interface 2. 0: 420 1: Reserved 2: 422 3: 444
			0x0	4:0	VC2_BIT WIDTH MINUS 1	Specifies the bit width – 1 for video streaming interface 2. I.e., value of 5'd23 indicates bit width of 24.
0x193	VIDEO_INTF3_COLOR_PATTERN	RO	0x0	31:13	RESERVED	Reserved.
			0xF	12:9	VC3_COLOR SPACE	Specifies the color space for video streaming interface 3. 0: RGB 1-4: Reserved 5: RAW 6: RGB565 7: YUV 8: YUVlegacy 9-15: Reserved
			0x0	8:7	VC3_COSITE	Specifies the co-siting of the chroma samples for video streaming interface 3.
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
						0: Reserved 1: Top and center (ie. YUV4208/10-bit Chroma Shifted Pixel Sampling) 2: Reserved 3: Center and center (YUV420 8/10-bit)
			0x1	6:5	VC3_SUBSAMPLING	Specifies the chroma sampling for video streaming interface 3. 0: 420 1: Reserved 2: 422 3: 444
			0x0	4:0	VC3_BIT WIDTH MINUS 1	Specifies the bit width – 1 for video streaming interface 3. I.e., value of 5'd23 indicates bit width of 24.
0x194 – 0x1AF	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x1B0	CORE_STATUS_IRQ_STATUS	RW1C	0x0	31:4	RESERVED	Reserved.
				3:0	VIDEO_INTF<ch>_STABLE	Stable bit of video streaming interface 0 to 3. When the IP detects a consistent width and height on the input video stream, it asserts this bit.
0x1B1 – 0x1B3	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.
0x1B4	CV2AXI_INTERRUPT_MASK_ENABLE	RW	0x0	31:4	RESERVED	Reserved.
				3:0	IRQ_EN_CORE_STATUS_INTF0	Interrupt enable registers allow you to selectively generate an interrupt at the output port for each error bit in the registers
continued...						

Address [9:0]	Register	Access	Default	Bit	Bit Name	Description
						0x1B0. Set to 0 to disable the interrupts.
0x1B5 – 0x3FF	RESERVED	N/A	N/A	N/A	RESERVED	Reserved.

6.2. CSI-2 Transmitter Register Map

Base address (`control_address[9:8]`) 0x0 accesses the MIPI CSI-2 transmitter control and status registers.) Base 0x1 accesses the `AXI2CV` registers. When you turn on **Enable the Control and Status Register** only registers in the address region from 0x100 to 0x1FF are accessible.

Table 29. CSI-2 Transmitter IP Registers

Address [9:0]	Register	RW	Default	Bit	Bit Name	Description
0x000	CORE_STATUS	RO	0x0	31:3	RESERVED	Reserved.
				2:0	MAXIMUM_LANES	Number of D-PHY lanes configured during IP generation minus 1, e.g. 4 lanes = 3.
0x001 – 0x03	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x004	GENERIC_SHORT_PACKET_STATUS	RO	0x0	31:5	RESERVED	Reserved.
				4:0	GSP_FIFO_USED	Generic short packet FIFO fill level.
0x005 – 0x00F	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x010	IRQ_STATUS	RW1C	0x0	31:3	RESERVED	Reserved.
				2	GSP_UNSUPPORTED_DT	Unsupported CSI-2 Data Type written to generic short packet FIFO.
				1	GSP_FIFO_FULL	Generic short packet FIFO is full and cannot accept more data.

continued...

Address [9:0]	Register	RW	Default	Bit	Bit Name	Description
				0	LINE_BUFFER_OVERFLOW	TX line buffer has overflowed, with data lost.
0x011 – 0x01F	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x020	IRQ_ENABLE	RW	0x0	31:3	RESERVED	Reserved.
				2	GSP_UNSUPPORTED_DT_ENA	Enable IRQ: Unsupported CSI-2 Data Type written to generic short packet FIFO.
				1	GSP_FIFO_FULL_ENA	Enable IRQ: Generic short packet FIFO is full and cannot accept more data.
				0	RESERVED	Reserved.
0x021 – 0x033	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x034	GENERIC_SHORT_PACKET_WRITE	RW	0x0	31:26	RESERVED	Reserved.
				25:0	GSP_DATA	Write a generic short packet to the transmit FIFO. Data should contain the 32-bit packet and a write to this register adds it to the FIFO for insertion.
0x035 – 0x0FF	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x100 – 0x14F	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x150	VIDEO_INTF0_STATUS	RO	0x0	31:1	RESERVED	Reserved.
				0	VC0_MODE_MATCH	Asserted when the input data resolution is matching
continued...						

Address [9:0]	Register	RW	Default	Bit	Bit Name	Description
						with the WIDTH and HEIGHT_F0/ HEIGHT_F1 registers for video streaming interface 0.
0x151	VIDEO_INTF1_STATUS	RO	0x0	31:1	RESERVED	Reserved.
				0	VC1_MODE_MATCH	Asserted when the input data resolution is matching with the WIDTH and HEIGHT_F0/ HEIGHT_F1 registers for video streaming interface 1.
0x152	VIDEO_INTF2_STATUS	RO	0x0	31:1	RESERVED	Reserved.
				0	VC2_MODE_MATCH	Asserted when the input data resolution is matching with the WIDTH and HEIGHT_F0/ HEIGHT_F1 registers for video streaming interface 2.
0x153	VIDEO_INTF3_STATUS	RO	0x0	31:1	RESERVED	Reserved.
				0	VC3_MODE_MATCH	Asserted when the input data resolution is matching with the WIDTH and HEIGHT_F0/ HEIGHT_F1 registers for video streaming interface 3.
0x154 – 0x15F	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x160	VIDEO_INTF0_WIDTH	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC0_WIDTH	Specifies the active video width(in
continued...						

Address [9:0]	Register	RW	Default	Bit	Bit Name	Description
						pixel unit) for video streaming interface 0.
0x161	VIDEO_INTF1_WIDTH	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC1_WIDTH	Specifies the active video width (in pixel unit) for video streaming interface 1.
0x162	VIDEO_INTF2_WIDTH	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC2_WIDTH	Specifies the active video width (in pixel unit) for video streaming interface 2.
0x163	VIDEO_INTF3_WIDTH	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC3_WIDTH	Specifies the active video width (in pixel unit) for video streaming interface 3.
0x164 – 0x16F	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x170	VIDEO_INTF0_HEIGHT_F0	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC0_HEIGHT_F0	Specifies the active video height (in line unit) of progressive video or interlaced video field 0 for video streaming interface 0.
0x171	VIDEO_INTF1_HEIGHT_F0	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC1_HEIGHT_F0	Specifies the active video height (in line unit) of progressive video or interlaced video field 0
continued...						

Address [9:0]	Register	RW	Default	Bit	Bit Name	Description
						for video streaming interface 1.
0x172	VIDEO_INTF2_HEIGHT_F0	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC2_HEIGHT_F0	Specifies the active video height (in line unit) of progressive video or interlaced video field 0 for video streaming interface 2.
0x173	VIDEO_INTF3_HEIGHT_F0	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC3_HEIGHT_F0	Specifies the active video height (in line unit) of progressive video or interlaced video field 0 for video streaming interface 3.
0x74 – 0x7F	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x180	VIDEO_INTF0_HEIGHT_F1	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC0_HEIGHT_F1	Specifies the active video height (in line unit) interlaced video field 1 for video streaming interface 0.
0x181	VIDEO_INTF1_HEIGHT_F1	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC1_HEIGHT_F1	Specifies the active video height (in line unit) interlaced video field 1 for video streaming interface 1.
0x182	VIDEO_INTF2_HEIGHT_F1	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC2_HEIGHT_F1	Specifies the active video height (in line unit)
continued...						

Address [9:0]	Register	RW	Default	Bit	Bit Name	Description
						interlaced video field 1 for video streaming interface 2.
0x183	VIDEO_INTF3_HEIGHT_F1	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC3_HEIGHT_F1	Specifies the active video height (in line unit) interlaced video field 1 for video streaming interface 3.
0x184 – 0x18F	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x190	VIDEO_INTF0_LINE_BLANK	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC0_LINE_BLANKING	Specifies the line blanking in pixels for video streaming interface 0.
0x191	VIDEO_INTF1_LINE_BLANK	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC1_LINE_BLANKING	Specifies the line blanking in pixels for video streaming interface 1.
0x192	VIDEO_INTF2_LINE_BLANK	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC2_LINE_BLANKING	Specifies the line blanking in pixels for video streaming interface 2.
0x193	VIDEO_INTF3_LINE_BLANK	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC3_LINE_BLANKING	Specifies the line blanking in pixels for video streaming interface 3.
0x194 – 0x19F	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x1A0	VIDEO_INTF0_FRAME_BLANK	RW	0x0	31:16	RESERVED	Reserved.
continued...						

Address [9:0]	Register	RW	Default	Bit	Bit Name	Description
				15:0	VC0_FRAME_BLANKING	Specifies the frame blanking in lines for video streaming interface 0.
0x1A1	VIDEO_INTF1_FRAME_BLANK	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC1_FRAME_BLANKING	Specifies the frame blanking in lines for video streaming interface 1.
0x1A2	VIDEO_INTF2_FRAME_BLANK	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC2_FRAME_BLANKING	Specifies the frame blanking in lines for video streaming interface 2.
0x1A3	VIDEO_INTF3_FRAME_BLANK	RW	0x0	31:16	RESERVED	Reserved.
				15:0	VC3_FRAME_BLANKING	Specifies the frame blanking in lines for video streaming interface 3.
0x1A4 _ 0x1AF	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.
0x1B0	VIDEO_INTF0_CONTROL	RW	0x0	31:2	RESERVED	Reserved.
				1	VC0_LINE_FRAMING_PKT_ENABLE	When set, enables insertion of MIPI line start and line end short packets for video streaming interface 0.
				0	INTERLACED	Specifies whether video is interlaced for video streaming interface 0.
0x1B1	VIDEO_INTF1_CONTROL	RW	0x0	31:2	RESERVED	Reserved.
continued...						

Address [9:0]	Register	RW	Default	Bit	Bit Name	Description
				1	VC1_LINE_FRAMING_PKT_ENABLE	When set, enables insertion of MIPI line start and line end short packets for video streaming interface 1.
				0	INTERLACED	Specifies whether video is interlaced for video streaming interface 1.
0x1B2	VIDEO_INTF2_CONTROL	RW	0x0	31:2	RESERVED	Reserved.
				1	VC2_LINE_FRAMING_PKT_ENABLE	When set, enables insertion of MIPI line start and line end short packets for video streaming interface 2.
				0	INTERLACED	Specifies whether video is interlaced for video streaming interface 2.
0x1B3	VIDEO_INTF3_CONTROL	RW	0x0	31:2	RESERVED	Reserved.
				1	VC3_LINE_FRAMING_PKT_ENABLE	When set, enables insertion of MIPI line start and line end short packets for video streaming interface 3.
				0	INTERLACED	Specifies whether video is interlaced for video streaming interface 3.
0x1B4 – 0xFF	RESERVED	N/A	N/A	31:0	RESERVED	Reserved.



7. MIPI CSI-2 IP User Guide Archive

For the latest and previous versions of this user guide, refer to [MIPI CSI-2 IP User Guide](#). If an IP or software version is not listed, the user guide for the previous IP or software version applies.

8. Document Revision History for MIPI CSI-2 IP User Guide

Document Version	Quartus Prime Version	IP Version	Changes
2026.08.10	26.1.1	4.0.0	- Added supported MIPI data rates information in the <i>Recommended Speed Grades for MIPI CSI-2 IP</i> topic. - Added <i>Licensing the IP</i> topic. - Added <i>IIP Image Property Values by Data Type</i> table in the <i>Receiver AXI4-Stream Output Video Interface Signals</i> topic. - Updated the description in the <i>Transmitter AXI4-Stream Input Passthrough Interface</i> topic. - Updated the description for MIPI D-PHY continuous clock parameter in the <i>IP Core Parameters</i> table. - Added <i>Timing Configuration</i> topic. - Updated <i>Troubleshooting</i> topic to add troubleshooting steps for the transmitter.
2026.05.01	26.1	3.1.0	- Updated the support level of Agilex 3 devices to preliminary in the <i>MIPI CSI-2 IP Device Family Support</i> table. - Updated the <i>MIPI CSI-2 IP Clocks</i> table: - Updated <code>axi4s_clk</code> clock description. - Added <code>slave_clock</code> and <code>ls_clk</code> clocks. - Added the <code>slave_clock</code> and <code>ls_clk</code> signals in the <i>MIPI CSI-2 IP Resets</i> table. - Added Separate clock for Control and Status Register interface (CSR) and MIPI DPHY continuous clock parameters in the <i>IP Core Parameters</i> table. - Updated Parameter Editor diagram. - Updated the description in the <i>Limitations and Known Issues</i> topic. - Updated RX PPI and Control and status register description in the <i>MIPI CSI-2 IP RX Functional Blocks</i> table. - Updated PPI and Control and status register block description in the <i>MIPI CSI-2 Transmitter Functional Blocks</i> table.
2025.03.30	25.1	2.0.0	- Added support for Agilex 5 D-series and Agilex 3 devices. - Added three new resource optimization parameters.
2024.10.07	24.3	1.3.0	- Updated <i>Limitations and Known Issues</i> - Added <i>Troubleshooting</i>.
continued...			

8. Document Revision History for MIPI CSI-2 IP User Guide

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Document Version	Quartus Prime Version	IP Version	Changes
2024.07.23	24.2	1.2.0	Corrected table headings in <i>MIPI CSI-2 Performance and Resources</i>
2024.07.08	24.2	1.2.0	• Updated <i>Performance and Resources</i> tables • Removed <i>sim/cadence</i> from <i>Generated Files</i> • Updated <i>Limitations and Known Issues</i> • Updated data types and scrambling support • Updated register maps.
2024.04.26	24.1	1.1.0	Initial release.